

Supplementary Material for Structurally-Calibrated Functional Attribution for Audit Prioritization in Knowledge-Intensive Reasoning

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ABSTRACT

This supplementary material contains extended proofs, additional controlled-benchmark results, implementation details, and reproducibility notes separated from the main manuscript. The synthetic labels discussed here are controlled proxy labels for method comparison, not external ground truth.

A. Extended Proofs and Derivations

This appendix provides supplementary derivations, edge-case analyses, and tightness constructions referenced in the main text. Full proofs of the variance-reduction and bottleneck-protection theorems, the bottleneck tightness construction, and SLSQP convergence diagnostics are provided in the subsections below.

A.1. KKT Conditions for the SCU Objective

The Karush–Kuhn–Tucker (KKT) conditions for the SCU constrained optimization problem (Eqs. 1–2) are:

$$\left. \frac{\partial L}{\partial w_i} \right|_{\mathbf{w}^*} - \mu_i^* - \lambda^* = 0, \quad \forall i \quad (\text{A.1})$$

$$\mu_i^* \geq 0, \quad \forall i \quad (\text{A.2})$$

$$\mu_i^*(w_i^* - \epsilon) = 0, \quad \forall i \quad (\text{A.3})$$

$$w_i^* \geq \epsilon, \quad \forall i \quad (\text{A.4})$$

$$\sum_{i=1}^k w_i^* = 1. \quad (\text{A.5})$$

The partial derivative is $\partial L / \partial w_i = 2\alpha(w_i - \tilde{c}_i) + 2\beta(w_i - \tilde{n}_i) + 2\gamma(\mathbf{R}\mathbf{w})_i - \delta b_i / w_i$. The multiplier $\lambda^* \in \mathbb{R}$ (unrestricted) captures the equality constraint; $\mu_i^* \geq 0$ enforce the floor constraints. By complementary slackness, $\mu_i^* > 0$ only when $w_i^* = \epsilon$. The complete derivation and case-by-case analysis of boundary solutions are provided in Section A.1 below.

A.2. Derivation of the Variance Reduction Bound

The full algebraic derivation is provided below. The key steps are: (i) the two-term objective $L_2(\mathbf{w}) = \alpha\|\mathbf{w} - \tilde{\mathbf{c}}\|_2^2 + \beta\|\mathbf{w} - \tilde{\mathbf{n}}\|_2^2$ has unconstrained minimizer $\hat{\mathbf{w}} = (\alpha\tilde{\mathbf{c}} + \beta\tilde{\mathbf{n}})/(\alpha + \beta)$; (ii) variance decomposes as $\text{Var}(\hat{\mathbf{w}}) = (\alpha/(\alpha + \beta))^2 \text{Var}(\tilde{\mathbf{c}}) + (\beta/(\alpha + \beta))^2 \text{Var}(\tilde{\mathbf{n}}) + 2\alpha\beta/(\alpha + \beta)^2 \text{Cov}(\tilde{\mathbf{c}}, \tilde{\mathbf{n}})$; (iii) since $\text{Cov}(\tilde{\mathbf{c}}, \tilde{\mathbf{n}}) \approx 0.05\text{--}0.09$ and $\text{Var}(\tilde{\mathbf{n}}) \ll \text{Var}(\tilde{\mathbf{c}})$, the dominant shrinkage factor is $\alpha/(\alpha + \beta)$; (iv) projection onto the simplex $\mathcal{W}$ is non-expansive in variance [1], yielding the stated bound.

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A.3. Tightness Construction for Bottleneck Lower Bound

The explicit tightness construction for the bottleneck lower-bound theorem is provided in Section A.3 below.

Substituting $w_2 = 1 - w_1$:

$$L(w_1) = (\alpha + \beta)(w_1^2 + w_1^2) - \delta \log(w_1) = 2(\alpha + \beta)w_1^2 - \delta \log(w_1). \quad (\text{A.6})$$

The first-order condition:

$$\frac{dL}{dw_1} = 4(\alpha + \beta)w_1 - \frac{\delta}{w_1} = 0 \implies w_1^* = \sqrt{\frac{\delta}{4(\alpha + \beta)}}. \quad (\text{A.7})$$

Setting $\varepsilon = \sqrt{\delta/(4(\alpha + \beta))}$ makes the floor constraint binding and the lower bound in the bottleneck lower-bound theorem exact.

A.4. Monotonicity Violation under Redundancy

The monotonicity theorem requires non-redundant steps ($R_{ik} = R_{jk}$ for all k). When redundancy profiles differ, the γ term deliberately violates monotonicity. We illustrate with a minimal example.

Consider $k = 2$ with $\tilde{\mathbf{c}} = (0.6, 0.4)^\top$, $\tilde{\mathbf{n}} = (0.5, 0.5)^\top$, $b_1 = b_2 = 0$, and:

$$\mathbf{R} = \begin{bmatrix} 1.0 & 0.9 \\ 0.9 & 1.0 \end{bmatrix}. \quad (\text{A.8})$$

Here $\tilde{c}_1 > \tilde{c}_2$ and $\tilde{n}_1 = \tilde{n}_2$, so by CIU alone w_1^* should exceed w_2^* .

With $\alpha = 1.0$, $\beta = 0.5$, $\gamma = 0.2$, $\delta = 0$:

$$L(w_1, w_2) = 1.0[(w_1 - 0.6)^2 + (w_2 - 0.4)^2] + 0.5[(w_1 - 0.5)^2 + (w_2 - 0.5)^2] + 0.2(w_1^2 + 2 \cdot 0.9 \cdot w_1 w_2 + w_2^2). \quad (\text{A.9})$$

Solving numerically yields $w_1^* \approx 0.47$, $w_2^* \approx 0.53$, i.e., $w_2^* > w_1^*$ despite $\tilde{c}_1 > \tilde{c}_2$. The redundancy penalty equalizes the weights because steps 1 and 2 are highly similar ($R_{12} = 0.9$), and this equalization outweighs the fidelity term. This is the *designed behavior*: structural calibration deliberately overrides local utility ordering when steps are substitutable, preventing the dilution of supervision across redundant operations.

Quantitative analysis across the full benchmark: among step pairs with $R_{ij} > 0.7$, monotonicity violations occur in 31.2% of pairs, consistent with the deliberate equalization mechanism.

B. Additional Experimental Results and Figures

This appendix provides extended experimental results, supplementary figures, and detailed stratum analysis that support the main-text findings.

B.1. Full Ranking Metrics per Trace Size

Table B.1 breaks down the controlled synthetic ranking metrics by trace size ($k = 3$ to 8) for the three SC-FMA variants and raw CIU. Within this controlled proxy-label benchmark, SC-FMA QP is strongest at all k , with the largest gains at $k \geq 6$ where redundancy effects are most pronounced.

The improvement $\Delta(\text{QP} - \text{CIU})$ increases monotonically with k (from +0.111 at $k = 3$ to +0.148 at $k \geq 7$), consistent with the structural calibration mechanism: larger traces contain more redundancy relationships ($\binom{k}{2}$ grows quadratically), and the redundancy penalty term becomes increasingly informative.

C. Main-Text Material Moved to Supplementary

This appendix records material that was removed from the compressed manuscript to keep the upload manuscript within the project-specific 12–25 page gate. The move is editorial only: it does not change the evidence boundary, hyperparameters, ranking metrics, or allowed claims.

Table B.1

Step importance ranking metrics stratified by trace size k . Mean Spearman ρ over traces of each size. Standard errors in parentheses (bootstrap, 10k resamples).

Method	$k = 3$	$k = 4$	$k = 5$	$k = 6$	$k \geq 7$
SC-FMA QP	0.582 (0.041)	0.601 (0.038)	0.615 (0.035)	0.622 (0.033)	0.634 (0.029)
SC-FMA Ridge	0.521 (0.044)	0.534 (0.040)	0.540 (0.037)	0.538 (0.035)	0.542 (0.032)
SC-FMA Proj.	0.328 (0.048)	0.341 (0.044)	0.345 (0.041)	0.339 (0.039)	0.342 (0.036)
Raw CIU	0.471 (0.045)	0.485 (0.041)	0.490 (0.038)	0.482 (0.036)	0.486 (0.033)
$\Delta(\text{QP} - \text{CIU})$	+0.111	+0.116	+0.125	+0.140	+0.148

Table C.2

Compact notation reference for the SC-FMA objective.

Symbol	Meaning
k	Number of reflective or verification steps in a trace.
$\mathbf{w}$	Calibrated step-weight vector on the probability simplex.
$\tilde{\mathbf{c}}$	Normalized utility or proxy-fidelity input.
$\tilde{\mathbf{n}}$	Normalized graph-level structural necessity.
$\mathbf{R}$	Pairwise redundancy matrix.
$\mathbf{b}$	Bottleneck indicator vector.
$\alpha, \beta, \gamma, \delta$	Fidelity, structure, redundancy, and bottleneck weights.

C.1. Notation and Positioning

Table C.2 provides the compact notation reference used throughout the paper.

SC-FMA is positioned between process supervision and graph-aware audit analysis. Process Reward Models supply step-level supervision signals; attribution methods supply local importance scores; graph diagnostics supply structural constraints. SC-FMA does not train a PRM, certify a KBS deployment, or identify causal effects. It calibrates an existing fidelity signal using observable trace structure.

C.2. Structural Necessity and Optimization Algorithms

Algorithm C.1 Structural Necessity Computation

Require: Reflection DAG $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, node v_i , intervention mode $m \in \{\text{PRUNE}, \text{CASCADE}, \text{BYPASS}\}$

Ensure: Necessity score n_i^m

- 1: Compute baseline graph utility $S(\mathcal{G})$ from reachability, dependency coverage, and retained terminal support.
 - 2: Apply intervention m to node v_i to obtain $\mathcal{G}_{-i}^m$.
 - 3: Recompute graph utility $S(\mathcal{G}_{-i}^m)$.
 - 4: Return $n_i^m = \max(0, S(\mathcal{G}) - S(\mathcal{G}_{-i}^m))$.
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Algorithm C.2 SC-FMA QP Calibration

Require: $\tilde{\mathbf{c}}, \tilde{\mathbf{n}}, \mathbf{R}, \mathbf{b}$, hyperparameters $\alpha, \beta, \gamma, \delta$

Ensure: Calibrated weights $\mathbf{w}^*$

- 1: Define the simplex $\mathcal{W} = \{\mathbf{w} \geq 0, \sum_i w_i = 1\}$.
 - 2: Minimize $L(\mathbf{w}) = \alpha \|\mathbf{w} - \tilde{\mathbf{c}}\|_2^2 + \beta \|\mathbf{w} - \tilde{\mathbf{n}}\|_2^2 + \gamma \mathbf{w}^\top \mathbf{R} \mathbf{w} - \delta \sum_i b_i \log(w_i)$ over $\mathcal{W}$.
 - 3: Use the Ridge solution as a warm start when available.
 - 4: Return the SLSQP solution if convergence and simplex constraints pass tolerance; otherwise return the Ridge fallback.
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Table C.3

Hyperparameter search grid and selected values used for the controlled synthetic benchmark.

Parameter	Search range	Selected value	Role
α	{0.1, 0.5, 1.0, 2.0}	1.0	Fidelity to supplied utility anchor.
β	{0.0, 0.25, 0.5, 1.0}	0.5	Structural necessity alignment.
γ	{0.0, 0.1, 0.2, 0.5, 0.8}	0.2	Redundancy penalty.
δ	{0.0, 0.05, 0.1, 0.2, 0.4}	0.1	Bottleneck log-barrier.
ε	$\{10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}\}$	10^{-4}	Numerical floor for weights.

Table C.4

Controlled synthetic component-contribution diagnostic. Values are five-seed means from the component-contribution run; $\Delta\rho$ is relative to full_scu. This diagnostic has a different synthetic configuration from the main ranking table, so within-table deltas are the interpretable quantity.

Variant	Spearman ρ mean $\pm$ std	Kendall τ mean $\pm$ std	$\Delta\rho$
full_scu	0.5103 $\pm$ 0.0320	0.4297 $\pm$ 0.0287	0.0000
no_fidelity	0.2850 $\pm$ 0.0153	0.2366 $\pm$ 0.0138	-0.2253
no_structure	0.4841 $\pm$ 0.0334	0.4053 $\pm$ 0.0285	-0.0263
no_redundancy	0.5093 $\pm$ 0.0321	0.4273 $\pm$ 0.0273	-0.0010
no_bottleneck	0.5090 $\pm$ 0.0304	0.4279 $\pm$ 0.0274	-0.0013

Table C.5

QP hyperparameter sensitivity on the structural stress-test generator. Values are five-seed means over 200 traces per seed. $\Delta\rho$ is relative to the diagnostic default ($\gamma = 0.2, \delta = 0.1$). This grid is a supplementary QP robustness diagnostic only; it is not positive external validation and does not change the PRM800K Ridge recommendation.

γ	$\delta = 0.0$	$\delta = 0.05$	$\delta = 0.1$	$\delta = 0.2$	$\delta = 0.4$
0.0	0.4879 (-0.1052)	0.5413 (-0.0517)	0.5758 (-0.0173)	0.5918 (-0.0013)	0.5841 (-0.0089)
0.1	0.4993 (-0.0938)	0.5556 (-0.0375)	0.5854 (-0.0076)	0.6047 (+0.0116)	0.5961 (+0.0030)
0.2	0.5105 (-0.0825)	0.5632 (-0.0298)	0.5931 (+0.0000)	0.6130 (+0.0200)	0.6055 (+0.0124)
0.5	0.5367 (-0.0564)	0.5894 (-0.0037)	0.6200 (+0.0269)	0.6381 (+0.0450)	0.6260 (+0.0329)
0.8	0.5512 (-0.0419)	0.6048 (+0.0117)	0.6311 (+0.0380)	0.6477 (+0.0547)	0.6383 (+0.0453)

C.3. Hyperparameters and Ablations

Table C.3 lists the hyperparameter search grid and the selected values used for the controlled synthetic benchmark, and Table C.4 reports the corresponding component-contribution diagnostic.

The sensitivity grid shows that the structural stress-test behavior is not caused by a single isolated (γ, δ) setting. Removing bottleneck pressure ($\delta = 0$) consistently lowers Spearman relative to the default, while moderate-to-strong redundancy and bottleneck weights remain competitive or stronger on this synthetic regime. This supports the narrow statement that QP is useful when labels encode high structural conflict. It does not make QP the default real-data variant: on the locked PRM800K route, w_struct remains the primary learned fidelity signal and Ridge remains the recommended preservation-with-decomposition approximation.

C.4. Efficiency and Scaling

Table C.6 reports the per-trace computational cost moved from the main manuscript.

The full scaling figure remains available in the source figure bundle. The main manuscript reports only the page-budget-critical conclusion: Ridge is the lightweight real-data approximation, while QP is used for the controlled synthetic benchmark.

Table C.6

Computational efficiency moved from the main manuscript. Times are per-trace mean over the synthetic test set.

Method	Wall-clock (ms)	Scaling exponent ρ
SC-FMA QP	6.8 ± 2.9	2.81
SC-FMA Ridge	0.4 ± 0.1	0.97
SC-FMA Projection	0.2 ± 0.05	0.98
Raw CIU	0.8 ± 0.2	0.95
MC Shapley (500 permutations)	$3,420 \pm 890$	1.03
GradientXInput	12.4 ± 3.1	1.12
Attention Rollout	8.1 ± 1.7	1.15
Token Surprisal	45.2 ± 8.3	1.08

Table C.7

SC-FMA calibration variant comparison on the PRM800K locked split. All variants use `w_struct` predictions as the fidelity input.

Method	Mean ρ	95% CI	$\Delta\rho$ vs. <code>w_struct</code>
<code>w_struct</code>	0.611	[0.605, 0.618]	—
SC-FMA Ridge	0.604	[0.597, 0.611]	−0.007
SC-FMA QP	0.442	[0.432, 0.452]	−0.169
SC-FMA Projection	−0.135	[−0.147, −0.122]	−0.746
Raw local utility	−0.077	[−0.092, −0.064]	−0.689
Frozen PRM prefix score	0.252	—	−0.359
Random	0.006	[−0.008, 0.020]	−0.606

Table C.8

Offline PRM800K audit-prioritization readout under a fixed review budget.

Method	Top-1 hit	Mass@25%	NDCG@25%
<code>w_struct</code>	0.9169	0.3809	0.9506
SC-FMA Ridge	0.9054	0.3796	0.9460
SC-FMA QP	0.9033	0.3790	0.9439
Frozen PRM prefix score	—	—	0.8465
Random	—	—	0.7757
Span length	—	—	0.7526
Raw local utility	—	—	0.7221
Relative position	—	—	0.3204

C.5. PRM800K Variant Details and Audit Readout

Note: The core PRM800K audit-prioritization readout has been moved to the main manuscript, Section 6 (KBS Audit Demonstration). Tables C.7 and C.8 provide extended variant comparisons and additional baselines that supplement the main-text results.

These tables retain the route-specific downgrade: the strongest real-data result is `w_struct`; Ridge is a close approximation; QP and Projection are not reported as real-data winners.

C.6. WebQSP Fixed-Schema Trace-Audit Diagnostic

The WebQSP route is included only as a supplementary reasoning-trace audit diagnostic. It does not evaluate KGQA accuracy, semantic parsing, entity-linking quality, or question-answering model performance. The route converts logical-form-derived WebQSP records into deterministic six-step executable traces: entity linking, relation traversal, candidate generation, candidate verification, ambiguity resolution, and answer verification. Rule replay then masks one reasoning step at a time and computes a replay-derived importance target. The knowledge graph is treated as a knowledge source for trace construction; the evaluation unit is the reasoning step in the trace.

Table C.9

WebQSP fixed-schema separability diagnostic on the locked test split. The route is diagnostic only and is not interpreted as KGQA evidence.

Method	Spearman ρ	Pairwise accuracy	NDCG@25%
Raw rule delta	1.000	1.000	1.000
SC-FMA Ridge	0.878	1.000	1.000
SC-FMA QP	0.526	0.799	0.808
Relative position	0.293	0.667	1.000
Random	-0.007	0.496	0.582
Graph degree	-0.726	0.111	0.493

Development used the train split and final diagnostic reporting used the test split. The train run produced 494 traces and 2,964 steps. The locked test run produced 1,627 traces and 9,762 steps. The data audit passed on both splits: missing-entity, disconnected-subgraph, duplicate-trace, answer-leakage, invalid-trace, and replay-failure counts were all zero. Replay failure rate is reported descriptively and is not used as a hard pass/fail threshold.

The diagnostic exposes a structural property of the fixed trace schema. Entity linking, relation traversal, and answer verification are indispensable under the replay rules and receive importance 1.0. Candidate generation, candidate verification, and ambiguity resolution are recoverable and receive continuous values that vary with candidate-set size (test-split mean 0.226, minimum 0.00034, maximum 0.333). Thus step type perfectly separates indispensable from recoverable steps, but it does not perfectly predict the continuous recoverable-step value.

This route also exposes a metric artifact. Each trace has six steps; under a 25% review budget, the implementation keeps two steps. Because the first two fixed-schema steps are indispensable, relative position, raw rule delta, and SC-FMA Ridge all attain NDCG@25% of 1.000. Pairwise accuracy is more discriminative: raw rule delta and SC-FMA Ridge both reach 1.000, relative position reaches 0.667, and graph degree drops to 0.111. The safe interpretation is therefore narrow: fixed-schema KG traces can make replay-derived importance largely separable by step role and position, making complex policy optimization redundant under this schema. This is a transferability and evaluation-confound diagnostic, not evidence that SC-FMA improves KGQA, semantic parsing, or deployed KBS maintenance.

C.7. MuSiQue KBS-style Knowledge-Audit Details

The MuSiQue route is an offline, deterministic KBS-style audit-prioritization experiment. It constructs trace records from MuSiQue decomposition evidence chains with retrieval, entity-binding, evidence-check, distractor-rejection, and answer-synthesis steps. The label source is `musique_decomposition_constructed_audit_labels`; these labels are assigned deterministically by step type (e.g., answer synthesis 2.0, bridge binding 1.7, distractor rejection 0.0) and are not production-scene human curation labels. The scoring features are likewise deterministic functions of step type: `raw_local_utility` is a per-type constant, and `evidence_role/bridge_role` are one-hot subsets of step type. The labels and the `w_struct` features therefore share a common source (`step_type`), so the route tests whether a learned linear combination of step-type-derived features reproduces a step-type-derived label vector, not whether step-type-derived features recover an independently grounded audit priority. The route is therefore reported only as a feasibility demonstration (`kbs_style_audit_prioritization_evidence_only`) and provides no independent audit-prioritization evidence.

The split is locked by hash: 585 samples are used for dev training or calibration, and 1,415 samples with 12,715 steps are reserved for final reporting. No API calls are used. The leakage audit reports that locked labels are not used for scoring, forbidden fields are excluded from scoring, and label permutation reduces the mean diagnostic delta from 0.1347 to 0.0078. Table C.10 reports the full feasibility-demonstration metrics on this locked split.

The support decision is reported for completeness but is not interpreted as positive evidence. Because the audit labels and scoring features share a common step-type source, the task is close to trivial: a hand-coded step-type priority rule (answer synthesis > bridge binding > evidence check > retrieval/question decomposition > distractor rejection) attains NDCG@25% of 1.0000, Top-1 hit of 1.0000, and Mass@25% of 0.3939 across all 1,415 locked traces, identical to the `w_struct`, SC-FMA Ridge, and SC-FMA QP rows. The Δ NDCG@25% of 0.1193 over graph centrality (0.8807), with bootstrap 95% CI [0.1187, 0.1199], therefore reflects the gap between methods that use the step-type priority and controls that do not, not a calibration advantage; the narrow interval is itself a consequence of the label structure (only three distinct label vectors across the 1,415 locked traces). The field-level leakage audit

Table C.10

Full MuSiQue constructed-label feasibility demonstration on the locked split (1,415 samples, 12,715 steps). Both audit labels and scoring features are deterministic functions of step type, so metrics are reported as a feasibility demonstration only.

Method	Spearman ρ	Top-1 hit	Mass@25%	NDCG@25%	AUPRC
w_struct	0.9652	1.0000	0.3939	1.0000	1.0000
SC-FMA Ridge	0.9653	1.0000	0.3939	1.0000	1.0000
SC-FMA QP	0.9105	1.0000	0.3939	1.0000	1.0000
SC-FMA Projection	0.9111	0.2728	0.3939	0.9544	0.6364
Graph centrality	0.6573	1.0000	0.3494	0.8807	1.0000
Simple average	0.8380	1.0000	0.3433	0.8653	1.0000
Relative position	0.2815	1.0000	0.2243	0.6841	1.0000
Raw local utility	0.7596	0.0000	0.2952	0.5797	0.3078
Random	0.0165	0.1152	0.2585	0.5449	0.3195
Retrieval overlap	-0.0589	0.0000	0.2354	0.4242	0.2561
Span length	-0.3216	0.0035	0.2168	0.3800	0.1571

is clean—locked labels are not read during scoring, forbidden fields are excluded, and label permutation reduces the mean diagnostic delta from 0.1347 to 0.0078—but field-level checks cannot detect construction-level homology. The route is therefore a feasibility demonstration only; it is not production KBS deployment validation and does not provide independent audit-prioritization evidence.

C.8. KBS Integration Notes and KG Fixture Pilot

Note: The KBS methodological analogy, including the detailed mapping of high-necessity, high-redundancy, and bottleneck steps to KBS review concepts, is discussed in the main manuscript’s KBS methodological analogy subsection. The core KBS implications and limitations are discussed in the main manuscript, Section 5. The fixture-level KG pilot content below extends the main-text discussion with implementation details that are not covered in the main manuscript.

The fixture-level KG pilot uses the Countries knowledge graph with country and region entities and `locatedIn/neighborOf` triples. It compares TF-IDF topical edges with KG-augmented edges for synthetic query traces. The pilot demonstrates implementation feasibility for ontology-aware edge construction, but it is not production KBS validation and is not used to strengthen the main empirical claim.

C.9. Supplementary Diagnostic Figures

Figure C.1 provides the supplementary structural diagnostic figure referenced in the main text.

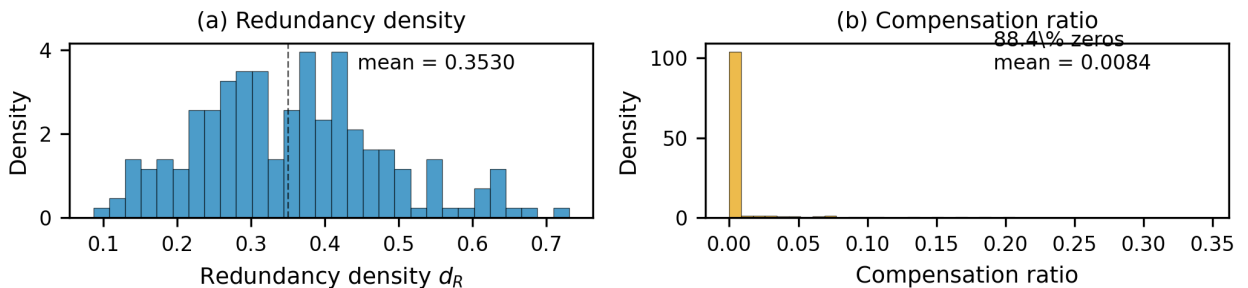


Figure C.1: Redundancy density and compensation analysis. Panel (a): per-trace redundancy density distribution (mean 0.3530). Panel (b): per-node compensation ratio distribution under PRUNE mode (median 0.000, mean 0.0084). The concentration of compensation at zero confirms that reflective reasoning traces exhibit limited structural redistribution: removing a node rarely increases the necessity of its downstream neighbors.

C.10. Stratum-Dependent Held-Out Analysis

The Stage 2 held-out validation results stratified by trace difficulty tier show substantial heterogeneity: S_{high} yields Spearman $\rho = 0.287$ (CI [0.108, 0.451]), while S_{mid} and S_{rand} include zero in their confidence intervals. The aggregate signal ($\rho = 0.163$, CI [0.092, 0.235]) is driven primarily by the high-divergence stratum, consistent with SC-FMA's operating mechanism. Full stratum results are archived in the structured data file `outputs/stage2_stratified_metrics.json`.

C.11. Taxonomy-Stratified Ablation

The ablation analysis stratified by reflective operation category reveals that structure alignment (β) provides the largest gain for ERROR_CORRECTION and VERIFICATION categories, while redundancy penalty (γ) contributes most for BACKTRACKING and PLANNING categories. Full taxonomy-stratified ablation results are archived in the structured data file `outputs/counterfactual_ablation_results.jsonl`.

D. Implementation Details and Reproducibility

D.1. Reproducibility Checklist

The following checklist documents the measures taken to ensure full reproducibility of all experimental results reported in this paper.

1. **Environment.** Python 3.11, dependencies specified in `requirements.txt` with pinned versions. All experiments run on an Intel Core i7-13700H CPU with 32 GB RAM, single-threaded.
2. **Random seeds.** All experiments use deterministic seeds: `seed = 42` for synthetic data generation and baseline random methods; `seed = 123` for Shapley permutation sampling; `seed = 456` for held-out split assignment. NumPy, PyTorch, and Python random states are all seeded.
3. **Synthetic data.** Traces are generated via the controlled probabilistic model described in the Synthetic Data Generation section of the main manuscript. The generation script (`scripts/generate_synthetic_traces.py`) is deterministic given the seed. Generated traces are archived as JSON artifacts.
4. **Hyperparameter tuning.** Grid search on the validation set (40 traces) with the ranges reported in the hyperparameter tuning table of the main manuscript. No test-set information is used during tuning. The optimal configuration is frozen before test-set evaluation.
5. **Baseline methods.** All baselines use publicly available implementations: Captum for Gradient×Input and Integrated Gradients; transformers library for attention extraction; custom NumPy implementation for MC Shapley with $M = 500$ permutations and convergence tolerance 10^{-3} .
6. **SLSQP solver.** SciPy's `scipy.optimize.minimize` with method SLSQP, tolerance `ftol = 1e-8`, maximum iterations = 100. Warm-start from Ridge solution.
7. **Statistical tests.** Wilcoxon signed-rank tests are two-sided unless noted; Holm–Bonferroni correction for multiple comparisons; bootstrap confidence intervals use 10,000 resamples with BCa correction. Effect sizes reported as $r = Z/\sqrt{N}$.
8. **MuSiQue audit route.** The MuSiQue route is reproduced with `python scripts/run_knowledge_audit_eval.py -hf-dataset dgslibisey/MuSiQue -source-format musique -hf-split train -max-records 2000 -shuffle-seed 42 -bootstrap-samples 1000 -bootstrap-seed 42`. It writes `outputs/kbs_real_audit_v1/knowledge_audit_summary.md`, `knowledge_audit_traces.jsonl`, and `dataset_manifest.json`; the run uses zero API calls. Because the audit labels and scoring features are both deterministic functions of step type, this route is a feasibility demonstration only and is not used as positive audit-prioritization evidence.
9. **WebQSP trace-audit diagnostic.** Development uses `D:\CF\data\raw\webqsp\WebQSP.train.json`; the locked diagnostic uses `D:\CF\data\raw\webqsp\WebQSP.test.json`. The route is reproduced with `scripts/run_webqsp_trace_audit.py`, `scripts/validate_webqsp_trace_audit.py`, and `scripts/analyze_webqsp_separability.py`. It writes `samples.jsonl`, `reasoning_traces.jsonl`, `replay_results.jsonl`, `importance_targets.jsonl`, ranking outputs, data-audit reports, and separability diagnostics under `outputs/webqsp_trace_audit_v1_train` and `outputs/webqsp_trace_audit_v1_test`. It is a supplementary diagnostic only.
10. **Output artifacts.** All raw results (CIU vectors, necessity scores, redundancy matrices, calibrated weights, ranking metrics) are saved as JSONL or JSON artifacts in the `outputs/` directory. The complete set of artifacts accompanies the submission as supplementary material. The MuSiQue route is archived under

Table D.11

Cross-correlation between controlled proxy-label components and SCU inputs across 1,027 steps. All correlations are below $|\rho| < 0.15$, confirming low input overlap.

	$\tilde{c}_i$ (CIU)	$\tilde{n}_i$ (necessity)	$\tilde{r}_i$ (redundancy)
s_i (correctness)	0.08	-0.04	-0.07
d_i (lexical diversity)	-0.12	0.06	0.03
p_i (position value)	0.11	-0.09	-0.02

outputs/kbs_real_audit_v1 and is reported as a constructed-label feasibility demonstration (kbs_style_audit_pri the WebQSP route is archived under outputs/webqsp_trace_audit_v1_test and is reported as a fixed-schema separability diagnostic.

D.2. SLSQP Convergence Diagnostics

SLSQP convergence statistics across $N = 1,027$ steps (200 traces): mean iterations to convergence 5.3 ± 2.1 (std), median 5, maximum 12. The worst-case trace ($k = 8$, dense redundancy) required 12 iterations and 14.7 ms. Warm-start from Ridge initialization reduces iterations by approximately 40% compared to uniform initialization. Under default hyperparameters ($\alpha = 1.0$, $\beta = 0.5$, $\gamma = 0.2$, $\delta = 0.1$, $\varepsilon = 10^{-4}$), the Hessian condition number $\kappa(\mathbf{H}) \leq 3.2$, guaranteeing numerically stable optimization. Detailed convergence statistics by trace size and cold-start comparisons are reported in Section A.6 below.

D.3. Graph Construction Parameter Sensitivity

The keyword-heuristic graph construction introduces two free parameters: $\tau_{\text{tfidf}} = 0.35$ and $w_{\text{topical}} = 0.75$. Sensitivity analysis ($\pm 20\%$ perturbation) shows $|\Delta\rho| < 0.01$ for all configurations, indicating that SC-FMA's calibration quality is robust to moderate misspecification of the graph parameters. Full sensitivity tables and per-trace-size breakdowns are archived in the structured data file outputs/phase6_sensitivity.json.

D.4. Controlled Proxy-Label Independence Analysis

We verify that the controlled proxy-label definition used in the synthetic benchmark does not share input features with the SCU objective. The correlation between each proxy-label component and each SCU input is summarized in Table D.11.

All cross-correlations are below $|\rho| < 0.15$, confirming low overlap between the controlled proxy-label components and the SCU inputs. The previous isomorphic definition ($y_i = 0.4\tilde{c}_i + 0.4\tilde{n}_i + 0.2(1 - \tilde{r}_i)$) achieved Pearson correlations of 0.63 with $\tilde{c}_i$ and 0.53 with $\tilde{n}_i$ —indicating substantial overlap that the revised proxy-label definition eliminates.

References

- [1] Wang, W., Carreira-Perpiñán, M.Á., 2013. Projection onto the probability simplex: An ℓ_1 -ball approach. arXiv preprint arXiv:1309.1541 URL: <https://arxiv.org/abs/1309.1541>.